Chapter 11, Problem 15PP

Problem

Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 220-V (rms), 60-Hz line.

Step-by-step solution

Step 1 of 8

Calculate the value of the power factor angle of the load before the capacitor is connected across the load.

$$\theta_1 = \cos^{-1}(0.85)$$
$$= 31.788°$$

Therefore, the value of the power factor angle θ_1 is, $31.788°$.

Step 2 of 8

Calculate the value of the apparent power S absorbed by the load before capacitor is connected across the load.

$$S_1 = \frac{Q_1}{\sin\theta_1}$$

Substitute $31.788°$ for θ_1 and 140 kVAR for Q_1.

$$S_1 = \frac{140\text{ k}}{\sin 31.788°}$$
$$= \frac{140\text{k}}{0.5268}$$
$$= 265.756\text{ kVA}$$

Therefore, the value of the apparent power S_1 is 265.756 kVA.

Step 3 of 8

Calculate the value of the real power P_1.

$$P_1 = S_1 \cos\theta_1$$

Substitute 265.756 kVA for S_1 and $31.788°$ for θ_1.

$$P_1 = (265.756\text{ k})\cdot\cos(31.788°)$$
$$= (265.756\text{ k})\cdot(0.85)$$
$$= 225.893\text{ kW}$$

Therefore, the value of the real power P_1 is 225.893 kW.

Step 4 of 8

Calculate the value of the power factor angle after the capacitor is connected across the load.

$$\begin{aligned}\theta_2 &= \cos^{-1}(1)\\ &= 0°\end{aligned}$$

Therefore, the value of the power factor angle θ_2 is $0°$.

Step 5 of 8

Calculate the value of the apparent power absorbed by the load after the capacitor is connected across the load.

$$S_2 = \frac{P_2}{\cos\theta_2}$$

Substitute $0°$ for θ_2 and 225.893 kW for P_2 since, no change is seen in the value of the real power.

$$\begin{aligned}S_2 &= \frac{225.893 \text{ k}}{\cos 0°}\\ &= \frac{225.893 \text{ k}}{1}\\ &= 225.893 \text{ kVA}\end{aligned}$$

Therefore, the value of the apparent power S_2 is 225.893 kVA.

Step 6 of 8

Calculate the value of the reactive power Q_2 after the capacitor is connected across the load.

$$Q_2 = S_2 \sin\theta_2$$

Substitute $0°$ for θ_2 and 225.893 kVA for S_2.

$$\begin{aligned}Q_2 &= (225.893 \text{ k})\cdot\sin(0°)\\ &= (225.893 \text{ k})(0)\\ &= 0 \text{ VAR}\end{aligned}$$

Therefore, the value of the reactive power Q_2 is 0 VAR.

Step 7 of 8

Calculate the value of the reactive power Q_C due to capacitor.

$$Q_C = Q_1 - Q_2$$

Substitute 0 VAR for Q_2 and 140 kVAR for Q_1.

$$\begin{aligned}Q_C &= 140 \text{ k} - 0\\ &= 140 \text{ kVAR}\end{aligned}$$

Therefore, the value of the reactive power due to capacitor Q_C is 140 kVAR.

Step 8 of 8

Calculate the value of the capacitance C

$$C = \frac{Q_C}{\omega V_{rms}^2}$$

Substitute $2\pi f$ for ω.

$$C = \frac{Q_C}{2\pi f V_{rms}^2}$$

Substitute $140\ \text{kVAR}$ for Q_C, $60\ \text{Hz}$ for f and $220\ \text{V}$ for V_{rms}.

$$\begin{aligned} C &= \frac{140\ \text{k}}{2\pi(60)(220)^2} \\ &= \frac{140\ \text{k}}{2\pi(60)(48400)} \\ &= \frac{140\ \text{k}}{18246.37\ \text{k}} \\ &= 7.673\ \text{mF} \end{aligned}$$

Therefore, the value of the capacitor C is, $\boxed{7.673\ \text{mF}}$.